

AVERY REYNA

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SKILLS

Languages: Python, TypeScript, JavaScript, Java

Frameworks & Libraries: React, Angular, AngularJS, React Native, D3.js, React Flow, Pandas, NumPy, Spark

Tools & Platforms: AWS (Lambda, S3, CloudWatch), Docker, New Relic, Git, GraphQL, Socket.io

WORK EXPERIENCE

Capital One

2023 – 2026

Software Engineer

McLean, VA

- Built first iteration of in-house monitoring system using **React**, **TypeScript**, and **Python** to replace Grafana dashboards, driving a **40% reduction** in third-party monitoring tool reliance by democratizing data quality insights across the organization
- Extended in-house metrics dashboard to all company users using **React** and **TypeScript**, building features including dynamic timeframe filtering, real-time threshold alerting, and granular cloud inventory tracking—improving anomaly detection across **500+ tracked resources**
- Developed custom **MITRE**-based security heatmap using **Angular** and **React** with saved layer panels and custom styling, streamlining threat discovery across the enterprise
- Implemented dynamic memory allocation for **S3** processing using **Python**, automatically scaling resources based on dataset size—saving **\$10K+ monthly** and preventing over-provisioning across **300+ datasets**
- Engineered parallel processing solution in **Python** to streamline configuration file parsing, reducing runtime from four minutes to under one minute (**75% improvement**) and decreasing system timeout issues by **90%**
- Led frontend development of **AWS** secrets management plugin using **AngularJS** and **Java**, reducing configuration time by **75%** across **200 AWS accounts** and streamlining vault operations for engineering teams
- Developed admin tooling with interactive data tables, inline editing, JSON view toggle, and batch delete capabilities using **React** and **TypeScript**—reducing metadata verification time by **50%** for cyber logging teams
- Built transmission monitoring integration into onboarding pipeline with **React/TypeScript** and detailed debug logging, enabling proactive error detection for **100+ quarterly datasets** and reducing support tickets by **40%**
- Upgraded file upload tool by translating **Python** validation script into **Java** and integrating real-time error feedback with **Angular**—slashing processing time by **2 hours** and eliminating all duplicate uploads
- Reduced **AWS Lambda** costs by **30%** through migration to Graviton processors, implementing **Docker** containerization to resolve NumPy dependencies while maintaining **99.9% reliability** via **CloudWatch** monitoring
- Spearheaded redesign of toxic access pairs plugin using **Angular**, resulting in a **40% reduction** in security vulnerabilities and **25% increase** in compliance with access control policies
- Integrated **GitHub Copilot** into **Jest** testing suites for internal applications, reducing development time by **30%** and increasing test coverage by **50%**
- Consolidated platform-wide styles into a global theme using **Vanilla Extract**, integrating it into a high-priority feature used by **12k+ employees** and advocating for adoption among engineering leadership
- Integrated **ML**-based Kai Slackbot into support channel with custom training dataset and regex-based conversation classification, deflecting **30%** of **50+ weekly support inquiries** with zero negative feedback

Self Adjoint Operator

2025 – Present

Design Engineer

Various Locations

- **Convoy** — Node-based data wrangling canvas for data visualization developers built on **React Flow**, designed around the workflows of practitioners who transform and visualize data in technical environments
- **Rerank** — Interactive text summarization tool built with **React** and **D3.js** that compares **TextRank**, **LexRank**, and **BART** algorithms with quality metrics analysis
- **Scrutinize** — Document annotation platform with **AI**-powered essay generation and interactive markup capabilities with structured editing codes for collaborative review workflows
- **Transparent** — Exploratory prototype investigating how **LLMs** can both generate user interfaces and explain associated code, probing mental models of AI-assisted development

RESEARCH EXPERIENCE

UC San Diego

2025 – Present

Independent Researcher

San Diego, CA

- Developed **Python**-based **GitHub GraphQL** scraper with parallel processing to extract **2.5k+ developer emails** from thousands of AI agent repositories for survey recruitment
- Led mixed-methods research studying developer-AI programming workflows, synthesizing findings into a research paper currently under submission to **ASE '26**

University of Kentucky

2026 – Present

Independent Researcher

Lexington, KY

- Designed the selected prototype for an **ML**-powered full-stack web application assessing global coup risk for **201 countries**, establishing the technical foundation adopted by a cross-functional research team
- Led a team of **6 CS students** in building new features, owning technical direction, task delegation, and code review while coordinating deliverables for a presentation to the **RAND Corporation**

PUBLICATIONS

Ruanqianqian (Lisa) Huang, **Avery Reyna**, Sorin Lerner, Haijun Xia, and Brian Hempel. “Professional Software Developers Don’t Vibe, They Control: AI Agent Use for Coding in 2025.” *arXiv Preprint*.

EDUCATION

University of Central Florida

Orlando, FL

Bachelor of Science – BS, Social Sciences

2023